APO-GABAPENTIN

Gabapentin capsules 300mg

Anti-epileptic

Clinical Pharmacology

Gabapentin exhibits antiseizure activity in mice and rats both in the maximal electroshock and in the pentylenetetrazol seizure models. Gabapentin is structurally related to the neurotransmitter GABA (gamma-aminobutyric acid) but does not interact with GABA receptors, it is not metabolized to GABA or to GABA agonists, and it is not an inhibitor of GABA uptake or degradation. Gabapentin at concentrations up to 100 µM did not demonstrate affinity for other receptor sites such as benzodiazepine, glutamate, glycine or N-methyl-D-aspartate receptors nor does it interact with neuronal sodium channels or L-type calcium channels.

The mechanism of action of gabapentin has not yet been established, however, it is unlike that of the commonly used anticonvulsant drugs.

The mechanism of the anticonvulsant action of gabapentin appears to be distinctly different from that of other antiepileptic drugs. Although structurally similar to GABA (gamma-aminobutyric acid), gabapentin at concentrations up to 1000 µM, did not bind to GABA receptors, it was not metabolized to GABA or a GABA agonist, and it did inhibit the uptake of GABA or its degradation by GABA-transaminase. Therefore, it does not appear to act through any known GABA mechanism, in contrast to the benzodiazepines, barbiturates, sodium valproate and other similar agents. Gabapentin (0.01 to 100 µM) did not interact with neuronal sodium channels or L-type calcium channels, in contrast to phenytoin, carbamazepine and sodium valproate which interact with these to promote the stability of excitable membranes. Finally, gabapentin (0.01 to 100 µM) did not interact with glutamate, glycine or N-methyl-D-aspartate (NMDA) receptors, in contrast to other drugs that have demonstrated anticonvulsant activity in animal models following interaction with these receptors. These neurophysiological findings indicate that gabapentin has a mechanism of action different from that of commonly used antiepileptic drugs.

<u>Pharmacokinetics</u>: Adults: Following oral administration of gabapentin, peak plasma concentrations are observed within 2 to 3 hours. Absolute bioavailability of a 300 mg dose of gabapentin capsules is approximately 59%. At doses of 300 and 400 mg, gabapentin bioavailability is unchanged following multiple dose administration.

Gabapentin elimination from plasma is best described by linear pharmacokinetics. The elimination half-life of gabapentin is independent of dose and averages 5 to 7 hours in subjects with normal renal function. Plasma gabapentin concentrations are dose-proportional at doses of 300 to 400 mg q8h, ranging between 1 and 10 µg/mL, but are less than dose-proportional above the clinical range (>600 mg q8h).There is no correlation between plasma levels and efficacy. Gabapentin pharmacokinetics are not affected by repeated administration, and steady-state plasma concentrations are predictable from single dose data.

Gabapentin is not appreciably metabolized in humans, is eliminated solely by renal excretion, and can be removed from plasma by hemodialysis.

Gabapentin does not induce or inhibit hepatic mixed function oxidase enzymes responsible for drug metabolism, does not interfere with the metabolism of commonly co-administered antiepileptic drugs, and is minimally bound to plasma proteins.

Food has no effect on the rate or extent of absorption of gabapentin.

In patients of epilepsy, gabapentin concentrations in cerebrospinal fluid are approximately 20% of corresponding steady-state trough plasma concentrations.

Indications

<u>Neuropathic Pain</u>

Apo-Gabapentin is indicated for the treatment of neuropathic pain.

<u>Epilepsy</u>

Apo-Gabapentin is indicated as adjunctive therapy in the treatment of partial seizures with or without secondary generalization in adults with epilepsy.

Dosage and Administration

<u>Neuropathic Pain</u>

Adults (over the age of 18)

Apo-Gabapentin should be titrated to a maximum dose of 1800 mg per day.

Titration to an effective dose can progress rapidly and can be accomplished over a few days by administering 300 mg once a day on day 1, 300 mg twice a day on day 2 and 300 mg three times a day on day 3, as described in Table 1.

Table 1: DOSING CHART - INITIAL TITRATION			
Dose	Day 1	Day 2	Day 3
900 mg	300 mg once a day	300 mg two times a day	300 mg three times a day

Thereafter, the dose can be increased using increments of 300 mg per day given in three divided doses to a maximum of 1800 mg per day. It is not necessary to divide the doses equally when titrating Apo-Gabapentin.

The maximum time between doses in a three times daily schedule should not exceed 12 hours. Gabapentin may be given orally with or without food.

If Apo-Gabapentin is discontinued, or the dose reduced or substituted with an alternative medication, this should be done gradually over a minimum of one week.

Elderly

Elderly patients may require dosage adjustment because of declining renal function with age (see Table 2).

<u>Epilepsy</u>

Apo-Gabapentin is recommended for add-on therapy in patients > 12 years. The effective dosage range was 900 to 1800 mg/day given in divided doses (3 times a day) using 300 mg capsules. Therapy may be initiated by titrating the dose as described below (see Table 1). Thereafter, the dose can be increased in three equally divided doses up to a clinically effective and tolerated dose. The first dose on Day 1 may be administered at bedtime to minimize potential side effects, especially somnolence, ataxia, fatigue and dizziness. Doses up to 2400 mg/day have been well tolerated in long-term open-label clinical studies. Doses of 3600 mg/day have also been administered to a small number of patients for a relatively short duration and have been well tolerated. APO-GABAPENTIN (gabapentin) is given orally with or without food.

Table 1: Titration schedule

Dose	Day 1	Day 2	Day 3
900 mg/day	300 mg OD	300 mg BID	300 mg TID

Daily maintenance doses should be given in three equally divided doses (see Table 2), and the maximum time between doses in a three times daily schedule should not exceed 12 hours. It is not necessary to monitor gabapentin plasma concentrations in order to optimize APO-GABAPENTIN therapy.

Further, as there are no drug interactions with commonly used antiepileptic drugs, APO-GABAPENTIN may be used in combination with these drugs without concern for alteration of plasma concentrations of either gabapentin or other antiepileptic drugs.

Dosage adjustment in elderly patients due to declining renal function and in patients with renal impairment or undergoing hemodialysis is recommended as follows (see Table 2).

Table 2: Maintenance dosage of APO-GABAPENTIN in adults with reduced renal function

Renal Function Creatinine Clearance (mL/min)	Total Daily Dose (mg/day)	Dose Regimen (mg)
>60	1200	400 three times a day
30 to 60	600	300 twice a day
15 to 30	300	300 once a day
<15	150	300 once daily every other day
Hemodialysis*	-	200 to 300**

* Loading dose of 300 to 400 mg

** Maintenance dose of 200 to 300 mg APO-GABAPENTIN following each 4 hours of hemodialysis

Mode of Administration
Oral Route

Contraindications
APO-GABAPENTIN (gabapentin) is contraindicated in patients who are demonstrated hypersensitivity to the drug or to any of the components of the formulation.

Warnings
Respiratory depression: Gabapentin has been associated with severe respiratory depression. Patients with compromised respiratory function, respiratory or neurological disease, renal impairment, concomitant use of central nervous system (CNS) depressants and the elderly might be at higher risk of experiencing this severe adverse reaction. Dose adjustments might be necessary in these patients.

Withdrawal Precipitated Seizures, Status Epilepticus: Antiepileptic drugs should not be abruptly discontinued because of the possibility of increasing seizure frequency.

Precautions
Gabapentin is not considered effective in the treatment of absence seizures and should therefore be used with caution in patients who have mixed seizure disorders that include absence seizures.
Laboratory Tests: Clinical trials data do not indicate that routine monitoring of clinical laboratory parameters is necessary for the safe use of gabapentin. Gabapentin may be used in combination with other commonly used antiepileptic drugs without concern for alteration of the blood concentrations of gabapentin of other antiepileptic drugs.
Effects on the Ability to Drive or Operate Machinery: Patients with uncontrolled epilepsy should not drive or handle potentially dangerous machinery. During clinical trials, the most common adverse reactions observed were somnolence, ataxia, fatigue and nystagmus. Patients should be advised to refrain from activities requiring mental alertness or physical coordination until they are sure that gabapentin does not affect them adversely.
Use in children: Systematic studies to establish safety and efficacy in children have not been performed.
Use in the elderly: As gabapentin is eliminated primarily by renal excretion, dosage adjustment may be required in elderly patients because of declining renal function (see Dosage and Administration)
Use in renal impairment: Gabapentin clearance is markedly reduced in this patient population and dosage reduction is necessary (see Table 2 in Dosage and Administration). Potential for an increase of suicidal thoughts or behaviors.

Drug Interaction
Antiepileptic agents: There is no interaction between gabapentin and phenytoin, valproic acid, carbamazepine, or phenobarbital. Consequently, gabapentin may be used in combination with other commonly used antiepileptic drugs without concern for alteration of the plasma concentrations of gabapentin or the other antiepileptic drugs.
Gabapentin steady-state pharmacokinetics are similar for healthy subjects and patients with epilepsy receiving antiepileptic agents.
Oral contraceptives: Co-administration of gabapentin with the oral contraceptive Norlestrin®; does not influence the steady-state pharmacokinetics of norethindrone or ethinyl estradiol.
Antacids: Co-administration of gabapentin with an aluminium and magnesium-based antacid reduces gabapentin bioavailability up to 20%. Although the clinical significance of this decrease is not known, co-administration of similar antacids and gabapentin is not recommended.
Probenecid: Renal excretion of gabapentin is unaltered by probenecid.
Cimetidine: A slight decrease in renal excretion of gabapentin observed when it is coadministered with cimetidine is not expected to be of clinical importance.

Use in pregnancy
This drug should only be used during pregnancy if the potential benefit to the mother justifies the potential risk to the fetus.

Use in lactation
Gabapentin is excreted in human milk. Because the effect on the nursing infant is unknown, caution should be exercised when gabapentin is administered to a nursing mother. Gabapentin should be used in nursing mothers only if the potential benefit outweighs the potential risks.

Adverse Reactions
Most common side effects are somnolence, dizziness, ataxia, fatigue, nystagmus and tremor. Others include headache, diplopia, nausea, vomiting, rhinitis, amblyopia, peripheral edema, coordination abnormal, depression and myalgia.
Body as a whole: Aesthenia, malaise or facial edema.
Cardiovascular system: Hypertension.
Digestive system: Anorexia, flatulence or gingivitis.
Hematologic and lymphatic system: Purpura most often described as bruises resulting from physical trauma.
Musculoskeletal system: Arthralgia.
Nervous system: Vertigo, hyperkinesia, parasthesia, anxiety, hostility and increased, decreased or absent reflexes.
Respiratory system: Pneumonia.
Urogenital system: Urinary tract infection.
Special senses: Abnormal vision.

Respiratory, thoracic and mediastinal disorders
Frequency 'rare': Respiratory depression

Post-marketing experience: Dysphagia

Symptoms and Treatment of Overdosage
Acute, life threatening toxicity has not been observed with gabapentin overdoses of up to 49 grams ingested at one time. In these cases, dizziness, double vision, slurred speech, drowsiness, lethargy and mild diarrhea were observed. All patients recovered with supportive care.
Gabapentin can be removed to hemodialysis. Although hemodialysis has not been performed in the few overdose cases reported, it may be indicated by the patients clinical state or in patients with significant renal impairment.
Reduced absorption of gabapentin at higher doses may limit drug absorption at the time of overdosing and, hence, reduce toxicity from overdoses.

Storage Conditions
Store below 30°C, in tightly closed and light resistant containers.

Product Description/Availability of Dosage Forms
APO-GABAPENTIN 300 mg: Each yellow opaque body with Yellow opaque cap; hard gelatin capsule, imprinted "APO 300", having a white to off-white powder fill, contains gabapentin equivalent of 300 mg. Available in PVC/PVDC blisters of 10x10's.

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